

LAB 1: AI-ASSISTED CITATION INTEGRITY AUDIT

Generate a Research Paper with AI, Then Audit Its Citations

AI Tools for Research • 2 Hours • Individual Activity • Topic IDs T1-T40

Student Detail	Fill in
Name	Harsh Sharma
Roll Number	MSE2026006
Programme / Department	MTECH I.T SOFTWARE ENGINEERING AUTOMATION
Topic ID	13
Full Assigned Topic	Planning-Quality Degradation in Long-Horizon Autonomous Research Agents
AI Tool Used	Google Gemini 3.6 Flash
Date	21/8/26

IMPORTANT: You will receive a topic from T1-T40. Your task is not to produce the “best” paper. Your task is to generate an AI-written research paper and systematically determine whether its citations can be trusted.

1. PURPOSE OF THIS LAB

Generative AI systems can produce research papers that look scholarly and may contain convincing references, author names, journal titles, DOIs, arXiv IDs, and in-text citations. However, the presence of a citation does not guarantee that the cited publication actually exists, that its metadata are correct, that the identifier belongs to that publication, or that the cited paper supports the statement made by the AI.

In this laboratory activity, you will test these issues yourself by generating a research paper and auditing a systematic sample of its references.

Question 1 - Citation Authenticity: Does the cited publication actually exist, and is it correctly identified?

Question 2 - Claim-Citation Support: If the publication is genuine, does it actually support the claim for which the AI cited it?

2. IMPORTANT RULES

- Use only the topic assigned to you by the instructor.
- Generate the paper in a fresh AI conversation.
- Do not tell the AI that you are conducting a citation audit or hallucination test.
- Do not intentionally ask the AI to generate fake references.
- Preserve the original AI-generated paper exactly as produced.
- Do not correct, regenerate, replace, or question references before saving the original output.
- For verification, do not accept another AI chatbot's statement as proof that a reference exists.
- Use scholarly databases, publisher pages, DOI/arXiv/PMID records, or other reliable scholarly sources.
- Record what you actually find, even if all references are genuine or all are problematic.

Most important principle: You are not being evaluated on whether the AI produces good or bad references. You are being evaluated on how systematically and correctly you audit them.

3. WHAT YOU WILL DO

Part	Activity
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A	Record your baseline perception of AI citations
B	Record the AI environment used
C	Generate and preserve an AI-written research paper
D	Create a reference inventory
E	Select 10 references systematically
F	Judge their plausibility before verification
G	Verify authenticity, metadata and identifiers
H	Determine whether genuine references support the AI-generated claims

4. RECOMMENDED 2-HOUR TIME PLAN

Time	Activity
0-10 min	Parts A-B: Baseline and AI environment
10-30 min	Part C: Generate and save research paper
30-40 min	Parts D-E: Inventory and select references
40-45 min	Part F: Pre-verification plausibility
45-80 min	Part G: Authenticity and metadata audit
80-110 min	Part H: Claim-citation entailment audit
110-120 min	Calculate scores, check worksheet and prepare submission

Do not spend excessive time formatting the AI-generated paper. The audit is the main activity.

PART A - BASELINE ASSESSMENT

Complete this section BEFORE opening the AI tool.

A1. How often do you use generative AI for academic work?

☐ Never ☐ Rarely ☐ Occasionally ☒ Weekly ☐ Almost daily

A2. Have you previously asked an AI system to generate references?

☐ Frequently ☐ Occasionally ☒ Once or twice ☐ Never

A3. Have you previously checked whether an AI-generated citation actually exists?

☐ Always ☐ Sometimes ☒ Rarely ☐ Never ☐ Not applicable

A4. Before this lab, how confident are you that citations generated by AI are generally reliable?

☐ 1 ☐ 2 ☐ 3 ☒ 4 ☐ 5 (1 = not at all confident; 5 = completely confident)

Important: Do not change this answer later, even if your opinion changes during the experiment.

PART B - RECORD YOUR AI ENVIRONMENT

Information	Your Response
AI tool used	Google Gemini 3.6 Flash
Model name	Google Gemini 3.6 Flash
Model/version shown	Google Gemini 3.6 Flash
Access tier	☒ Free ☐ Paid/Pro ☐ Institutional ☐ Not sure
Web browsing available?	☒ Yes ☐ No ☐ Not sure
Was web browsing actually used?	☒ Yes ☐ No ☐ Not sure
Research/Deep Research mode used?	☒ Yes ☐ No
Date/time generated	21/8/26

Important: If the model/version is not displayed, write NOT SHOWN. Do not guess the model version.

PART C - GENERATE THE AI RESEARCH PAPER

C1. Open a Fresh AI Conversation

Do not tell the AI that this is a citation audit, that you are testing hallucinations, or that you expect fabricated references. We want to observe normal AI research-paper generation behaviour.

C2. Use the Following Common Prompt

Replace only [YOUR ASSIGNED TOPIC].

Generate a scholarly research paper on the topic: "[YOUR ASSIGNED TOPIC]". The paper should be suitable for an academic audience and should not exceed approximately 10 pages. Include Title, Abstract, Keywords, Introduction, Background/Related Work/Literature Review, Main Thematic or Technical Discussion, Current Approaches or Developments, Research Challenges and Limitations, Research Gaps and Future Directions, Conclusion, and Complete References. Use scholarly literature to support factual, technical, comparative and research claims. Provide in-text citations throughout the paper. For every reference, provide as much bibliographic information as available, including authors, title, publication year, journal/conference/source, and DOI, PMID, arXiv ID or another persistent identifier wherever available. Use genuine scholarly sources. Do not intentionally fabricate references.

C3. Preserve the Original Output

- ☐ Save the complete AI-generated paper
- ☐ Save the complete reference list
- ☐ Take screenshot(s) showing the generated output/references
- ☐ Save conversation/export/share evidence if available

STOP: Do not ask the AI to correct or verify anything yet. Your original output is the experimental evidence.

PART D - PAPER PROFILE AND REFERENCE INVENTORY

Paper title: Planning-Quality Degradation in Long-Horizon Autonomous Research Agents

Measure	Value
Approximate pages	14
Approximate word count	3733
Number of sections	7
Total number of references	21
Approximate in-text citation occurrences	50
Did AI warn you to verify references?	☐ Clearly ☐ Vaguely ☒ No

D1. Reference Inventory

Classify every reference once using the best available identifier in this priority order: DOI -> PMID -> arXiv ID -> URL only -> No persistent identifier. The category totals must equal the total number of references in the AI-generated paper.

Reference Type	Count
DOI available	1
PMID available, no DOI	0
arXiv ID available, no DOI/PMID	18

URL only	0
No persistent identifier supplied	2
TOTAL	21

Examples

Category	Example	How to Count
DOI available	Kather, J. N., et al. (2019). Predicting survival from colorectal cancer histology slides using deep learning. PLoS Medicine. DOI: 10.1371/journal.pmed.1002730	DOI available
PMID available, no DOI	A biomedical paper for which the AI provides PMID: 12345678, but no DOI is supplied/found	PMID available, no DOI
arXiv ID available	Yao, S., et al. ReAct: Synergizing Reasoning and Acting in Language Models. arXiv:2210.03629	arXiv ID available
URL only	World Health Organization. Colorectal cancer. Available at: https://... with no DOI, PMID or arXiv ID	URL only
No persistent identifier	Goodfellow, I., Bengio, Y., & Courville, A. (2016). Deep Learning. MIT Press.	No persistent identifier if none is supplied

Important: Absence of a DOI does not mean a reference is fake. An identifier being present also does not prove authenticity; it must be checked.

PART E - SELECT 10 REFERENCES FOR AUDIT

You are not allowed to choose only suspicious-looking references. This prevents selection bias.

If the bibliography contains 10 or fewer references: audit all references.

If it contains more than 10 references: select the first 3 references, the last 3 references, and 4 references approximately evenly distributed through the middle.

Example: If there are 28 references, a reasonable sample could be 1, 2, 3, 8, 13, 18, 23, 26, 27, 28.

Audit Slot	Original Bibliography Reference No.
R1	1
R2	2
R3	3
R4	6
R5	9
R6	13
R7	15
R8	19
R9	20
R10	21

Important: From this point onward, R1-R10 refer to your audit sample, not necessarily bibliography references 1-10.

PART F - PLAUSIBILITY TEST

Do this BEFORE searching for any reference.

Look only at the citation as generated by the AI. Ask yourself: “If I had not verified this citation, how convincing would it look to me?”

Score	Meaning
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1	Obviously suspicious
2	Somewhat suspicious
3	Unsure
4	Looks genuine
5	Completely convincing

Ref.	Initial Impression	Plausibility 1-5	Predict Genuine?
R1	Realistic arXiv record with complete scholarly metadata.	5	Yes
R2	Specific GitHub repository and project title looked credible.	5	Yes
R3	Plausible recent arXiv citation, but less familiar and therefore uncertain.	3	Yes
R4	Specific arXiv identifier and detailed bibliographic information looked credible.	5	Yes
R5	Specific arXiv identifier and plausible author/title combination looked credible.	5	Yes
R6	Well-known topic and exact arXiv identifier looked credible.	5	Yes
R7	Known agent paper with a convincing NeurIPS/arXiv citation.	3	Yes
R8	Specific OpenReview ID and established paper title looked credible.	5	Yes
R9	Specific arXiv identifier and plausible recent AI-science paper looked credible.	5	Yes
R10	Specific arXiv identifier and plausible recent scholarly title looked credible.	5	Yes

Example: A citation with realistic authors, a well-known journal, complete volume/pages and a DOI-like identifier may look highly convincing. You might record Plausibility = 5 and Predict Genuine = Yes. You are recording your pre-verification belief, not the truth.

Critical rule: Never change Part F after completing Part G. Your incorrect predictions are valuable experimental observations.

PART G - CITATION AUTHENTICITY AND METADATA AUDIT

G1. What Are You Checking?

- Does the publication exist?
- Is the title correct?
- Are the authors correct?
- Is the year correct?
- Is the journal/conference/source correct?
- Are volume, issue and pages correct, if supplied?
- Does the DOI/arXiv/PMID belong to this publication?

G2. Verification Procedure

- ☐ Persistent identifier: DOI / arXiv / PMID
- ☐ Publisher or official repository
- ☐ Google Scholar
- ☐ Semantic Scholar / OpenAlex
- ☐ Crossref
- ☐ PubMed or another discipline-specific database
- ☐ Exact-title search

[!] Author + title-keyword search

Not acceptable as sole evidence: “I asked another AI and it said the paper exists.” AI can help you search, but final verification must rely on scholarly or publisher evidence.

G3. Assign One A-E Code

Code	Classification	Meaning
A	VERIFIED	Publication exists and important metadata are correct.
B	WRONG METADATA	Publication exists, but one or more bibliographic details are incorrect.
C	FRANKENSTEIN	Real-looking pieces are combined, but that specific publication does not exist.
D	FABRICATED	No reliable scholarly evidence that the publication exists.
E	IDENTIFIER MISMATCH	DOI/arXiv/PMID is invalid or resolves to a different publication.

G4. Examples

A: AI gives a paper and all details match the publisher record.

B: The paper exists, but the AI gives the wrong publication year or journal.

C: The authors are real, the journal is real, and the topic is plausible, but no such paper with that title/authorship combination exists.

D: You cannot find the paper or a credible scholarly record after systematic searching.

E: The DOI supplied by AI opens an entirely different paper or does not resolve.

G5. Record Your Evidence

Ref.	Publication Found?	Title Match?	Authors Match?	Year Match?	Venue Match?	Identifier Match?
R1	Y	Y	Y	Y	Y	Y
R2	Y	Y	Y	Y	Y	NA
R3	Y	Y	Y	Y	Y	Y
R4	Y	Y	Y	Y	Y	Y
R5	Y	Y	Y	Y	Y	Y
R6	Y	Y	Y	Y	Y	Y
R7	Y	Y	N	Y	Y	Y
R8	Y	Y	Y	Y	Y	Y
R9	Y	Y	Y	Y	Y	Y
R10	Y	Y	Y	Y	Y	Y

Ref.	Verification Source(s)	Identifier Checked	What You Found
R1	arXiv:2601.22311; official arXiv record	arXiv:2601.22311	Paper exists and title, authors, year and arXiv identifier match.
R2	GitHub: RUC-NLPIR/Awesome-Long-Horizon-Agents; official repository page	RUC-NLPIR/Awesome-Long-Horizon-Agents URL	Repository exists under RUC-NLPIR and the cited title/source match the repository.
R3	arXiv:2506.01372; official arXiv record	arXiv:2506.01372	Paper exists and title, authors, year and arXiv identifier match.
R4	arXiv:2606.31651; official arXiv record	arXiv:2606.31651	Paper exists and title, authors, year and arXiv identifier match.
R5	arXiv:2509.09677; official arXiv record	arXiv:2509.09677	Paper exists and title, authors, year and arXiv identifier match.
R6	arXiv:2310.04406; official arXiv record	arXiv:2310.04406	Paper exists and title, authors, year and arXiv identifier match.
R7	NeurIPS 2023 paper page; arXiv:2303.11366	arXiv:2303.11366	Paper exists and venue, year, title and identifier match, but the citation omits Edward Berman from the author list.
R8	OpenReview forum 6LNTSrJjBe; official OpenReview record	6LNTSrJjBe	OpenReview record exists for the cited paper; title, authors, 2023 submission date, source and identifier match.
R9	arXiv:2603.17216; official arXiv record	arXiv:2603.17216	Paper exists and title, authors, year and arXiv identifier match.
R10	arXiv:2607.26064; official arXiv record	arXiv:2607.26064	Paper exists and title, author, year and arXiv identifier match.

G6. Final Classification

Ref.	Final Code	One-Line Reason
R1	A	Verified: publication exists and the cited metadata match the official arXiv record.
R2	A	Verified: the cited GitHub repository exists and the repository title and source match.
R3	A	Verified: publication and cited arXiv metadata match the official record.
R4	A	Verified: publication and cited arXiv metadata match the official record.
R5	A	Verified: publication and cited arXiv metadata match the official record.
R6	A	Verified: publication and cited arXiv metadata match the official record.
R7	B	Article exists, but the citation omits coauthor Edward Berman from the author list.
R8	A	Verified: the OpenReview record, title, authors, 2023 submission date and identifier match.

R9	A	Verified: publication and cited arXiv metadata match the official record.
R10	A	Verified: publication and cited arXiv metadata match the official record.

Good reason: B: Article exists, but AI reported 2022 instead of 2020.

Poor reason: B: Details wrong.

PART G - SCORING

Count your final classifications: A = 9 B = 1 C = 0 D = 0 E = 0

Total audited: N = A + B + C + D + E = 10

Code	Points
A	4
B	3
C	1
D	0
E	1

Authenticity Points = $4A + 3B + C + E$

Authenticity Score = $[(4A + 3B + C + E) / (4N)] \times 100$

My calculation: $4(9) + 3(1) + 1(0) + 0(0) + 1(0) = 39$ points

Authenticity Score = $39 / (4 \times 10) \times 100 = 97.5 / 100$

Worked example: If A=6, B=2, C=1, D=1, E=0, then points = $4(6)+3(2)+1(1)=31$; maximum = 40; Authenticity Score = $31/40 \times 100 = 77.5$.

Score	Interpretation
90-100	Excellent authenticity
75-89	Good, minor concerns
60-74	Moderate citation-authenticity risk
40-59	High citation-authenticity risk
Below 40	Very high citation-authenticity risk

G7. Prediction Accuracy

Compare your Part F Genuine Yes/No predictions with the Part G results.

Prediction Accuracy = $(\text{Correct genuine/fake predictions} / \text{Total predictions}) \times 100$

Correct predictions = 10 Total predictions = 10 Prediction Accuracy = 100 %

FINAL LAB 1 RESULTS

Measure	Your Result
Total references in AI paper	21
References deeply audited	10
A: Verified	9
B: Wrong metadata	1

C: Frankenstein	0
D: Fabricated	0
E: Identifier mismatch	0
Authenticity Score	97.5 /100
Prediction Accuracy	100 %

FINAL INTERPRETATION

1. Did professional-looking citations necessarily turn out to be genuine?

No. The citations looked professional, but one audited reference contained incorrect author metadata. A convincing citation format did not guarantee perfect bibliographic accuracy.

This cannot be established from the bibliography alone. Claim-citation support must be checked against the exact claims and source content in the original AI-generated paper.

☐ The most serious issue found was wrong metadata in Reference 15: the cited paper is genuine, but the citation omits coauthor Edward Berman.

The key lesson is to verify both source authenticity and claim support independently. A valid DOI, arXiv ID, or URL does not by itself prove that all citation details or the cited claim are correct.

SUBMISSION INSTRUCTIONS

Submit one PDF: Lab1_RollNumber_Name

Example: Lab1_IEC2026001_RahulSharma

STUDENT DECLARATION

I confirm that I generated the research paper using the recorded AI environment, preserved the original output before conducting the audit, followed the prescribed reference-selection procedure, and personally verified the evidence reported in this worksheet. I have not altered my pre-verification predictions after seeing the verification results.

Student Name: Harsh Sharma

Roll Number: MSE2026006

Date: 21/8/26

CENTRAL LESSON OF LAB 1

Does the source exist? + Does the source actually support the claim?

AI Generation -> Reference Inventory -> Systematic Sampling -> Prediction -> Authenticity Verification -> Claim-Support Verification -> Quantitative Scores -> Submission